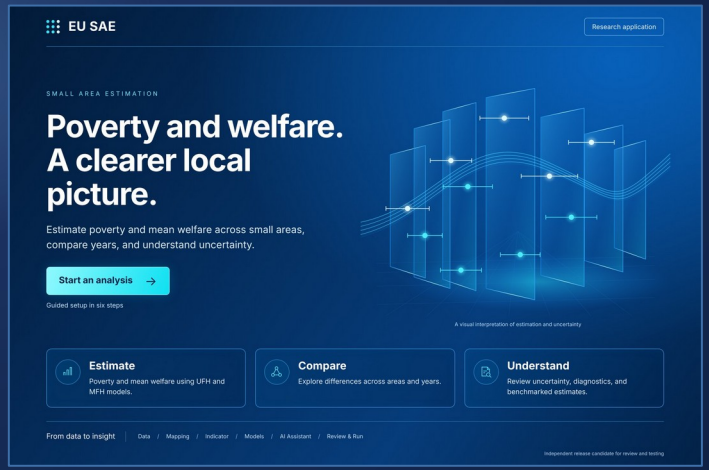
EU SAE WIZARD & DASHBOARD · 2026

A Practical Guide

to EU_SAE_application_package

Small Area Estimation of poverty and welfare indicators - from setup to outputs.



Front page of the EU SAE app (wizard and dashboard)

Wizard 5.14 · Pre-release for review

Open Start_Here to choose a launcher

Optional: AI with Claude or ChatGPT

Prepared by Nobuo Yoshida

What you'll learn

Four parts: setup, the complete wizard, worked examples and optional AI guidance.

01 Setup	Install R, download the app and prepare the required and supplementary datasets.
02 The complete wizard	Follow all six steps, from loading data to checking readiness, running the analysis and finding outputs.
03 Worked examples	Apply the wizard to year-specific poverty, anchored poverty and real mean welfare.
04 AI Assistant (optional)	Set up an approved Claude or ChatGPT provider and review the AI guidance.

OVERVIEW

What is the EU SAE application?

Two interfaces over one analysis engine for small area estimates of poverty and mean welfare.

Two entry points

Use the **wizard** (six guided steps) or the **classic dashboard** (one sidebar). Both **load** survey, auxiliary and geometry data, **fit** Fay–Herriot models (UFH and MFH), **compare** them with maps and tables, and can **add** AI-assisted interpretation.

```
Launchers for every system: Start_Here/
```

Check readiness
Inputs and diagnostics, before any model

UFH
Univariate Fay–Herriot: one model per year

MFH
Multivariate Fay–Herriot: years modelled jointly

Compare
UFH vs MFH maps and precision

What you can estimate

Three indicator families, all from the same survey-plus-auxiliary inputs.

INDICATOR 1

Poverty rate

FGT(0) · headcount



Share of the population below the poverty line — the standard poverty rate.

INDICATOR 2

Gap & severity

FGT(1) and FGT(2)



Depth of poverty (average shortfall) and severity (more weight on the poorest).

INDICATOR 3

Mean welfare

Income or consumption



Population-weighted mean welfare (optional log scale), in current or constant prices; changes reported in %.

Key terms

Used throughout the guide. Units matter: welfare is per person.

Term	Meaning
Domain	The small area you estimate for (e.g., province); same ID in all three files
Direct estimate	The survey-only estimate for a domain, before any model
UFH / MFH	Fay–Herriot models: one year at a time / both years jointly
PSU	Primary sampling unit: the cluster or area drawn at the first sampling stage
Welfare	Income or consumption per person (per capita or equivalised), not the household total
Weight × household size	How many people each household record represents
RMSE, CV	RMSE = √MSE; CV = RMSE ÷ estimate (×100 when shown as a percentage)

PART 01

Setting up your computer



Install R, download and launch the app, then prepare your data files.

Prerequisites at a glance

R and the app are required. Pandoc supports report generation; AI keys are optional.



R

R 4.2 or later (your approved version)

`cran.r-project.org`



EU SAE app

Wizard and dashboard, from GitHub Releases

`eu-sae-application/releases`



AI keys

Optional · Claude or ChatGPT

`see Part 4 of this guide`



Report generation

Pandoc 2.17+ supports HTML and Word reports. The app finds it or downloads a copy online.

`RStudio users: see slide 8`

Install R and check report tools

R runs the app. Pandoc generates HTML and Word reports.

1. Go to cran.r-project.org
2. Choose Download R for Windows, macOS or Linux
3. Run the installer with its default options
4. RStudio users: update RStudio Desktop if Pandoc is missing or too old.



Report generation

RStudio bundles Pandoc. Update to the latest stable version compatible with your computer, then restart the wizard. Alternative: standalone Pandoc 2.17+.

Download the app

Pre-release 5.2.0-rc.6 (wizard 5.14) is the file EU_SAE_520_w5n.zip on GitHub Releases.

1. Under Assets of the pre-release, download EU_SAE_520_w5n.zip from github.com/Noboyoshida227/eu-sae-application/releases — not the "Source code" archives.
2. Windows: right-click the ZIP → Properties → tick Unblock (if shown) → OK, then Extract All. macOS: double-click the ZIP.
3. Extract to any convenient folder. You get EU_SAE_520_w5n (5.2.0, wizard 5.14).
4. Open EU_SAE_520_w5n. Launchers are in Start_Here. Example data: Data/simulated/ and Data/Spain/.

```
EU_SAE_520_w5n/
├── Start_Here/
├── app.R + app_wizard.R
├── R/ + scripts/ + tests/
├── Data/  (examples)
├── docs/
├── outputs/  (empty)
├── LICENSE + NOTICE
└── release_manifest.csv
```

Launch the app

Open Start_Here and double-click the launcher for your system. If macOS blocks it: System Settings › Privacy & Security › Open Anyway (macOS 14 and earlier: right-click › Open).

```
Windows: Start_Wizard.bat     macOS: Start_Wizard.command
Linux: bash Start_Wizard.command
```



First launch

Missing R packages install first (internet needed). Then the front page opens in your browser.



Click Start an analysis

The wizard opens at Step 1 — Data.



Classic dashboard

For the alternative interface, use Start_Dashboard.bat or Start_Dashboard.command.

Three required files

You select all three in wizard Step 1 with Browse; they can sit in any folder.

DATASET 1

Survey

Household microdata

One row per household: year, domain ID, PSU, household weight, household size and per-person welfare (per capita or equivalised, not the household total). Optional: strata.

DATASET 2

Auxiliary

Subnational aggregates

One row per domain and year (long format): domain ID, year and candidate covariates, covering both analysis years.

DATASET 3

Geometry

Geographic boundaries

Polygons for the target domains, with a domain ID that matches the survey and auxiliary files.

Data formats are very flexible

Each input accepts many common formats — the app reads them automatically.

SURVEY

Household data

Any table format

RDS/RData, CSV, TSV, TXT, DAT, Stata, SPSS, SAS, Parquet, Feather and Excel. Person-level file? Add a household-size column of 1s.

AUXILIARY

Area covariates

Any table format

The same table formats as the survey: RDS/RData, CSV/TSV/TXT/DAT, Stata, SPSS, SAS, Parquet/Feather, and Excel.

GEOMETRY

Boundaries

Any GIS format

ESRI shapefile .zip (with valid .shp/.shx/.dbf), RDS/RData, GeoPackage (.gpkg), GeoJSON/JSON, KML, and GML.

Supplementary datasets

Add these files when the chosen analysis or benchmarking needs them.

CPI database	Optional. Needed only to convert current-price welfare to constant prices; not needed if welfare is already real. Type CPI values in Step 3, or load a file here (one row per year: year and CPI).
Population data	Optional; used for benchmarking. Domain totals by year, long or wide. Without it, the app uses the sum of weight × household size.
Benchmark targets	Optional; loaded in Step 4. Same indicator and the same prices as the survey welfare you upload. Under CPI conversion, supply current-price targets; the app converts them.

CPI and population files are loaded in Step 1. Years and prices are explained at the start of Part 3.

PART 02

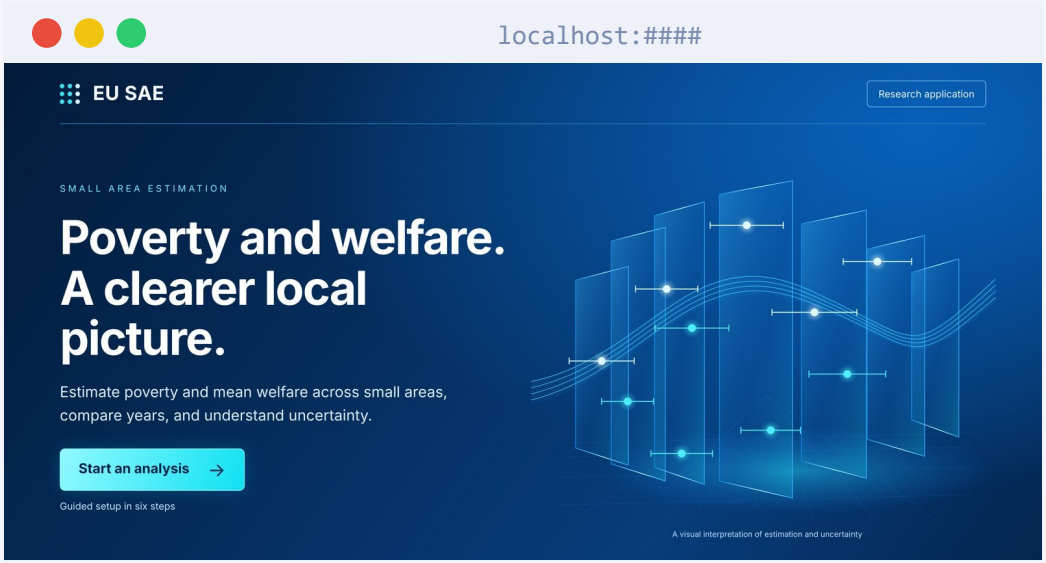
The wizard, step by step

Six steps from data to run. The classic dashboard offers the same controls in one sidebar.



Start here — the front page

The first screen in both interfaces. Click Start an analysis to open the workspace.

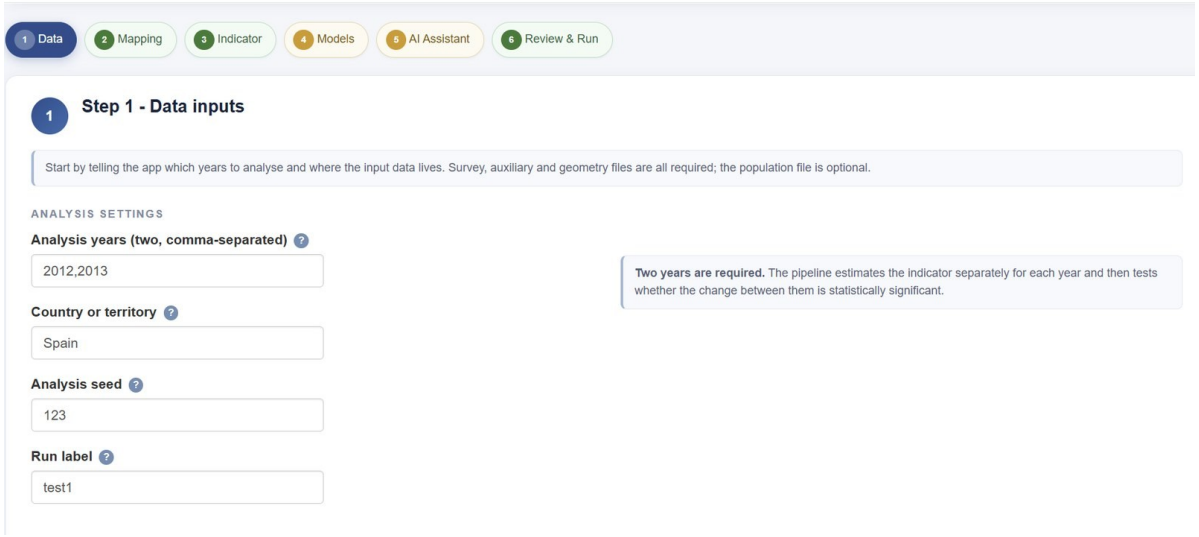


What you'll find

- Wizard: six steps on a clickable breadcrumb · Dashboard: one sidebar
- Check Data Readiness, then Run Analysis
- Preflight — live input checks (wizard Step 6)
- Data Readiness — filled by Check Data Readiness (Step 6)
- Status — progress, run folder and output locations

The wizard at a glance

Every one of the six steps shares this layout. Step 1 shown.



Breadcrumb
where you are, and what is still outstanding

Guidance banner
what this step is for, in one line

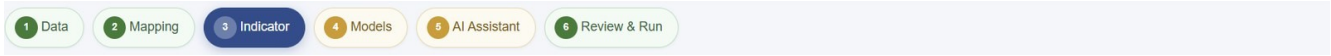
Form column
the inputs; help panels sit to the right

Back / Next
at the bottom of each step; nothing is locked

Three things to watch

The parts of the screen that tell you whether the run will work.

Breadcrumb - green is complete, amber has an outstanding item, blue is the current step. Clickable.



Validation — green confirms the step; red lists what to fix (here: grouped benchmarking is on but no variable is chosen).



Navigation — Next always works. Only Run Analysis is gated: it runs after Check Data Readiness passes without errors.



Step 1 — Data

Example survey years 2022 and 2023. Set the analysis years, then select the files.

Setting	Example
Analysis years	2022,2023
Country or territory	Simulated example
Analysis seed	123
Run label	example_2022_2023

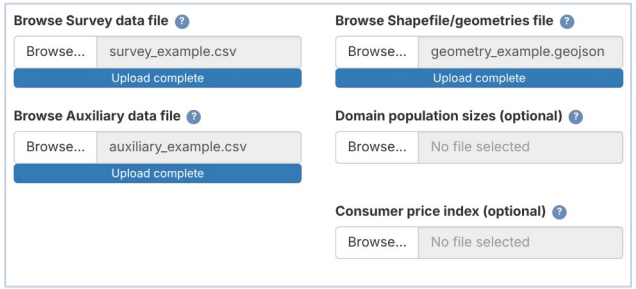
Analysis years

Two survey years, separated by a comma.

Country, seed and run label

Country appears in the report. The seed controls reproducibility. The run label names the archived outputs.

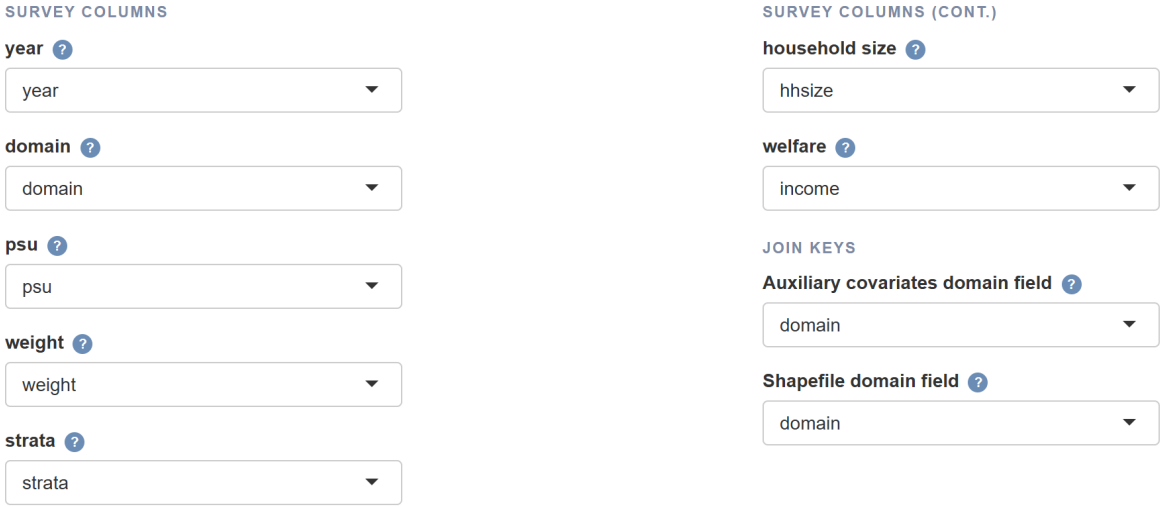
Below the settings: the input files



Survey, auxiliary and geometry files are required.
Population and CPI files are optional. CPI can also be typed in Step 3.
Simulated data: 2022–2023.

Step 2 — Variable mapping

Tell the app which column holds which variable.



strata: select the sampling-stratum column when available, otherwise leave blank. welfare: per person. weight: household weight. Match domain IDs across files. Choose the poverty-line column in Step 3. Red = column name not found.

Step 3 — Indicator

Choose what to estimate and confirm the welfare and price basis.

Indicator	Choose Poverty (FGT) or Mean welfare. For headcount rates, choose FGT(0) under Poverty measure.
Poverty lines	Poverty line source: Column in data (choose the column here) or Numeric value (one per analysis year). Match the welfare's currency, period and units.
Mean-welfare prices	Default: current prices, no conversion (changes include inflation). For real changes, convert with a price index, or choose constant prices already.

Part 3 shows the Step 3 settings for each worked example.

Step 4 — Covariates and PSU consistency

Every Step 4 option has a default. Check these two before you run.

Covariates

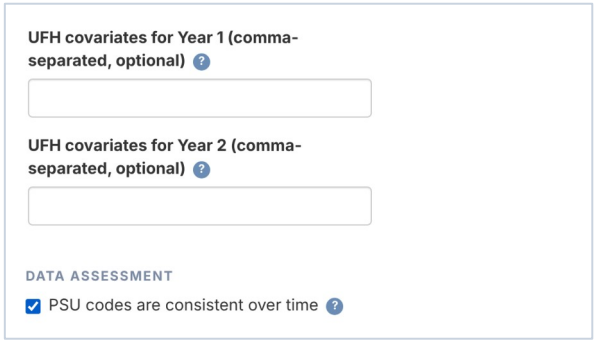
One comma-separated list per year, for UFH and for MFH. Blank: stepwise selection picks from all numeric auxiliary columns.

LASSO

Off (default): a list you type is a fixed model. On: LASSO screens the list as a candidate pool before stepwise selection.

PSU consistency

Under Data assessment, tick **PSU codes are consistent over time** only if the same PSU codes identify the same sampling units in both years. It is recorded with the run settings and does not change the estimates.



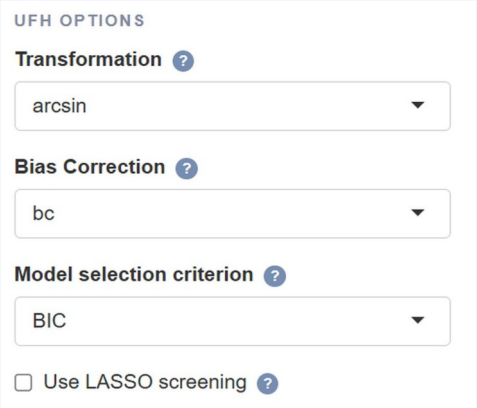
Wizard Step 4 (simulated example data)

Pipeline steps at the top of Step 4 choose which of UFH, MFH and Comparison run; all three are on by default.

UFH options

Defaults are listed first. Covariates and LASSO are on the previous slide.

Option	Values	What it does
Transformation	`FGT(0): arcsin / no` `welfare: log / no`	FGT(0): arcsin for bounded rates. log models the arithmetic area mean.
Bias Correction	`arcsin: bc / none` `log: bc_sm / none`	UFH log: native emdi Slud–Maiti correction. none omits correction.
Model selection criterion	`BIC (default) / AIC`	Criterion for stepwise selection.
Variance option (UFH)	`sm_out / sm_all / direct`	With no transformation: sm_out smooths missing or < 0.001 variances. sm_all smooths all. direct uses survey variances.



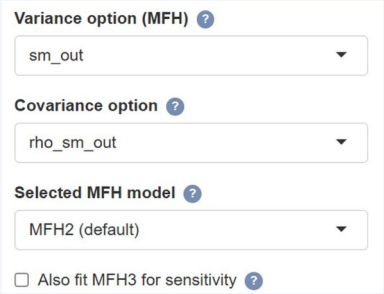
With arcsin, fill missing/zero variances only. With log, use sm_out.

Bootstrap replicates defaults to 500. More draws reduce simulation noise, not model bias.

MFH options

MFH2 is the default reported model. Covariates and LASSO are on slide 21.

Option	Values	What it does
Model selection criterion	`AIC (default) / BIC`	AIC favours predictive fit. UFH uses BIC.
Variance / Covariance option	`sm_out / sm_all / direct`	Available with log. Covariance choices follow the variance option (default: rho_sm_out).
Transformation (MFH)	`log (default) / no`	For mean welfare. Bias Correction (MFH): Conditional lognormal / none.
Selected MFH model	`MFH2 / MFH1 / MFH3`	MFH2 default. MFH3 starts a checked variance-structure comparison.
Bootstrap replicates	`500 (default)`	UFH, MFH and benchmarking. Check successful fits and simulation noise.



Model controls

MFH1: independent year effects MFH2: stationary AR(1), the default
MFH3: time-specific innovation variances, retained only with usable fit and supporting contrast
Zero variance or correlation at its limit: change significance is provisional. See slide 49.

Benchmarking (optional)

Matches population-weighted estimates to targets. Valid zero-variance MFH fits remain eligible.

1 · TURN ON

Apply benchmarking

Check Apply benchmarking. The Benchmarking level control and optional target database then appear.

2 · CHOOSE LEVEL

Benchmarking level

Choose National, or Grouped by a survey variable and select the grouping variable (for example region or NUTS2). Use survey-derived or uploaded targets.

3 · ADD TARGETS

Target file (optional)

Optional file: a year column and a target column (for example target or benchmark), plus a group column if grouped. Same prices as the survey welfare. Without it, direct estimates are the targets.

Step 5 — AI Assistant (optional)

Off by default. Nothing is sent anywhere unless you opt in here.

Two consent boxes before any key

tick Enable AI Assistant and the acknowledgement, then paste a Claude (sk-ant-…) or OpenAI (sk-…) key

Language

AI text comes back in the language you pick — 25 options

Where it lands

labelled blocks in the HTML and Word reports, written during Run Analysis — so enable AI before you run

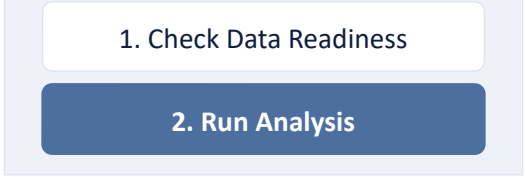
Keys and providers

see Part 4; keys are never stored on disk

Two-stage launch — check, then run

Review your data before the pipeline starts. Catch typos, missing years, and mapping issues early.

Wizard Step 6 · Dashboard sidebar



Stage 1 — review

Stage 2 — run the pipeline

Click 1, review, then click 2.

1. Check Data Readiness

Loads inputs; checks variables, years and mappings; fills the Data Readiness tab. No modelling yet.

2. Run Analysis

Starts only after the check passes with no ERROR. Change an input? Check again. Runs in the background; Stop ends a run.

Why two stages? You see what the app sees before committing to a long run - catch typos, missing domains, missing household size, or unexpected years early.

Preflight tab — inputs at a glance

Updates as you change inputs. Checks that the app can run: folder, packages, years and files.

Project root
`Working directory looks correct (the package folder)`

rmarkdown
`Package 'rmarkdown' is available for report rendering.`

Analysis years
`Two analysis years detected: 2022, 2023`

Spatial dependency
`Package 'sf' is available.`

Input data files
`Survey: available | Auxiliary: available | Geometry: missing`

What to look for

Green OK: the check passed

Amber Check: needs attention

Fix see Recommended actions below the checks

Variable mappings and data quality are checked by Check Data Readiness (Data Readiness tab).

Data Readiness - diagnostics

Populated when you click Check Data Readiness. Review warnings and errors before Run Analysis.

01

Year variable present

Both datasets include a year column.

02

Requested years available

Survey and auxiliary data include the two requested years.

03

Covariate summary

Mean, SE, correlation and its p-value; most significant first.

04

Domain/key alignment

Checks domain/year keys and geometry domains across all files.

05

Missing direct estimates

Flags empty domain-year cells.

06

National indicator

Population-weighted national poverty or mean welfare for selected years.

Where results live

Current outputs are in outputs/. Each run also has an archived copy under app_runs/.

outputs/final_report.html + .docx

Same results in both. Word text and tables are editable.

outputs/figures/

Maps, diagnostics, change and RMSE plots (.png).

pov_comparison_detailed.xlsx

In outputs/data/. Direct, UFH and MFH estimates with MSE, RMSE, CV (same name for mean welfare).

outputs/data/

Also EU_SAE_results.xlsx and change_estimate_comparison.xlsx.

outputs/tables/

CSV summaries: readiness, fit checks, covariance repairs and bootstrap diagnostics.

app_runs/…/app_config.yml

The run's saved settings, to reload later.

Note Each run overwrites outputs/. Edit a copy of the Word report, or use the archived copy in app_runs/.

Every-run checklist

Check data, numerical fit, bootstrap diagnostics and sensitivity before reporting.

SET UP · ONE TIME

Install once. Prepare files (Part 1).

1. Install R 4.2 or later
2. Download EU_SAE_520_w5n.zip from Assets
3.

```
Start_Here/
Windows: Start_Wizard.bat
macOS:
Start_Wizard.command
```

IN THE APP · EVERY RUN

Complete the six steps, then review the evidence.

1 Data	Years and required files
2 Mapping	Variables and survey weights
3 Indicator	Poverty line or welfare prices
4 Models	Covariates and model options
5 AI Assistant	Optional: consent, key and language
6 Review & Run	Check readiness, then run
Results	Open outputs/final_report.html

Current results: outputs/. Saved copy: app_runs/<timestamp>_<run_label>/outputs/.

PART 03

Worked examples



Three analyses, each through the six wizard steps of Part 2. Start with years and prices.

Years and prices: five terms

Survey, auxiliary, population and target files use survey years. CPI inputs depend on income years, the price base year and the CPI format. Geometry uses domain IDs.

Term	Meaning	Example
Survey (analysis) year	Typed in Step 1; keys the survey, auxiliary, population and target files	2022, 2023
Income reference year	Year the incomes refer to; CPI values are entered for these years (EU-SILC: usually the year before)	2021, 2022
CPI reference year	Where the index equals 100; it does not change results	2015 = 100
Price base year	Year whose prices are used to report real welfare	2021 prices
Anchor year	Year whose poverty line is held fixed in real terms	2021 line: 10,000

Example: CPI 2021 = 110.0, CPI 2022 = 114.4. Welfare 10,920 × 110.0 / 114.4 = 10,500 in 2021 prices.

The CPI reference year (2015) cancels out. Only the ratio between two years matters.

EXAMPLE 1 OF 3

Poverty headcount rates

Using the appropriate poverty line for each year.

Next: poverty-line inputs, indicator settings, model choices and interpretation.

Year-specific poverty lines

The threshold follows the poverty definition chosen for each year.

Measure	The population-weighted share of people whose welfare is below that year's poverty line.
Required line	One line per analysis (survey) year, in the same currency, accounting period and welfare units as the survey.
EU at-risk-of-poverty threshold	60% of the national median equivalised disposable income. Supply it per survey year; in EU-SILC it usually reflects the previous year's incomes.

A change in the rate can reflect changes in both welfare and the poverty threshold.

Poverty-specific inputs and settings

The survey and poverty line must describe the same welfare concept.

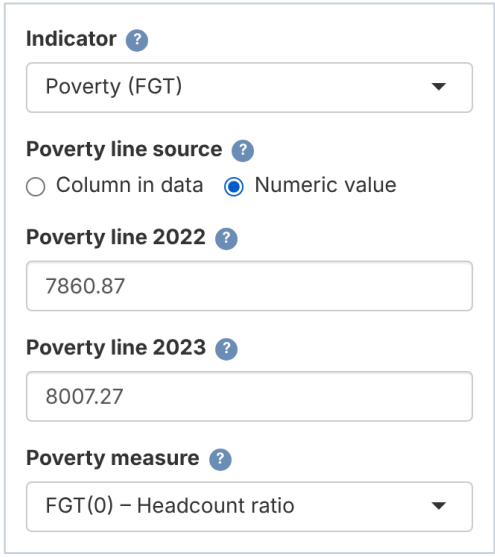
Welfare and years

Use the intended per-person or equivalised welfare measure. Supply a poverty line appropriate to each year and to the chosen poverty definition.

Line source

In Step 3, Poverty line source: **Column in data** (lines vary across records or areas) or **Numeric value**: one box per analysis (survey) year, as shown. Then choose Poverty (FGT) and FGT(0).

For a household file, person weights are survey weight × household size.



Survey years 2022–2023. Incomes refer to the year before.

Poverty models and interpretation

Use the supplied year-specific lines throughout estimation and interpretation.

Models (Step 4) For the standard setup, use UFH arcsin with bias correction and MFH2. Review covariates, variance choices and PSU consistency.

Survey year (Step 1)	Income year	Typed in Step 3	Headcount
2022	2021	Poverty line 2022: 7860.87	20.0%
2023	2022	Poverty line 2023: 8007.27	18.0%

Lines match slide 35. Illustrative headcounts: change −2.0 percentage points (−10% relative).

If benchmarking is enabled, targets must use the same poverty definition and lines.

EXAMPLE 2 OF 3

Anchored poverty headcount rates

EU application: keep the poverty standard fixed in real terms.

Supply the anchored lines: one number per survey year in Step 3, or a survey column.

A poverty line fixed in real terms

Keep the anchor-year threshold’s purchasing power constant over time.

Poverty-line source	Use official anchored thresholds where available. Check the anchor year and income reference year, and match the welfare definition and units.
Two inputs	Numeric value: type each survey year’s line in Step 3. Column in data: an anchored_povline column, when lines vary across records.
Step 3	Choose Poverty (FGT) and FGT(0), then Numeric value (one line per survey year) or Column in data (select anchored_povline).

Benchmarking domain estimates to national or regional targets is a separate choice.

Anchored lines for each survey year

Type these in Step 3 or store them in a survey column. Use official thresholds where available.

Survey year	Income reference year	Anchored line
2022	2021	10,000
2023	2022	10,400
2024	2023	10,608

Illustration: the supplied thresholds keep a 2021 poverty standard fixed in real terms.

Previous-year incomes: survey 2022 uses the 2021 threshold, and survey 2023 uses the 2022 threshold.

Settings for supplied anchored lines

The app uses the anchored thresholds you supply: typed in Step 3 or in a survey column.

Input

Numeric value: type 10000 (2022) and 10400 (2023) in Step 3. Or use an anchored_povline survey column.

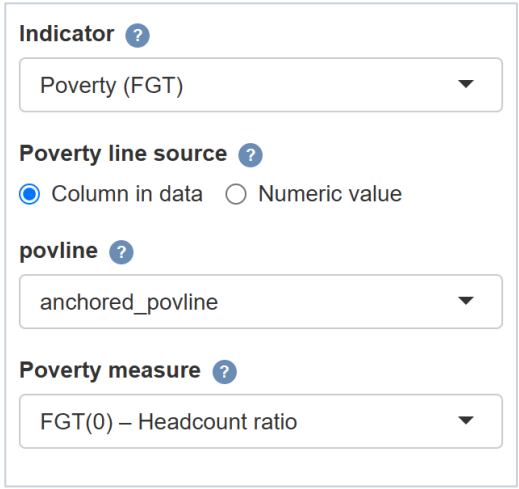
Step 3 · Indicator

Choose Poverty (FGT) and FGT(0). Poverty line source: Numeric value, or Column in data with anchored_povline. No CPI conversion.

Price consistency

Current-price welfare: use each income year's line in current prices. Welfare already in 2021 prices: the line is 10,000 in every year.

Check that the welfare and anchored line use the same prices and welfare units.



Shown: Column in data. The Numeric value boxes are as on slide 35.

Anchored-poverty checks and interpretation

Check the supplied thresholds before running the standard poverty workflow.

Before the run	Verify the source, anchor year, income years, currency and units. Check each year's line in the Step 6 summary or your column.
Comparable estimates	Keep the poverty definition, welfare units and model settings consistent when comparing years. The purchasing power of the poverty line stays fixed.
Results	Report anchored headcount rates and changes in percentage points. If benchmarking, use targets based on the same anchored poverty line.

Next: real mean welfare (Example 3).

EXAMPLE 3 OF 3

Real mean welfare

Estimate average welfare in a common year’s prices.

Route A: welfare already in real prices · Route B: current-price welfare + CPI

Two routes to real mean welfare

Estimate the population-weighted arithmetic mean in a common year’s prices.

Route A · real welfare	Per-person income or consumption for each household, already in constant prices of one year.
Route B · welfare + CPI	Per-person welfare in current prices plus CPI values (typed or from a file). The app deflates before estimating.
Input and output	Household records with per-person welfare. The app estimates Direct, UFH and MFH mean welfare by domain and year.

Both routes are available. The default is current prices, no conversion — change it in Step 3.

Welfare already in constant prices

The survey's per-person welfare is already adjusted for prices.

Survey welfare	Per-person welfare already in constant prices. Both survey years share the price base, welfare definition and currency.
Mapping (Step 2)	Map the real welfare variable as welfare. Retain the survey weights, household size and survey-design mappings from the general walkthrough.
Step 3 · Indicator	Choose Mean welfare. Under **Welfare in the survey data is in**, choose **constant (real) prices already**. The price year is optional (labels only).

No CPI conversion is applied. Continue to the mean-welfare model and results slides.

Current-price welfare plus CPI values

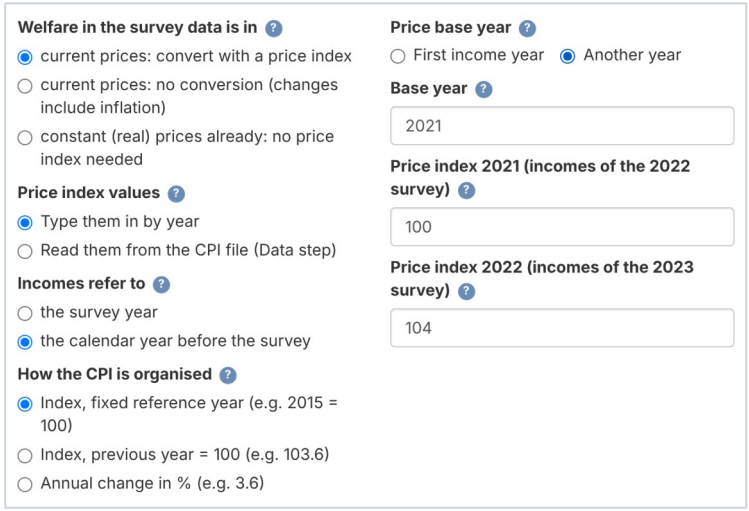
The app converts each household's welfare, then estimates real mean welfare.

CPI values — Type them by year in Step 3 (default) or read them from a CPI file loaded in Step 1.

Step 3 · Prices — Mean welfare → **current prices: convert with a price index**.

Step 3 · Years — Set **Incomes refer to**, **How the CPI is organised** and **Price base year**, then fill the CPI years the app lists.

The price base year determines the units of real welfare. It need not be the CPI reference year.



Simulated example: 2022–2023 surveys, incomes of the year before, base year 2021

Three accepted CPI formats

CPI file: one row per year (columns year and CPI). Use one national annual-average series.

How the CPI is organised (format key, not file layout)	2021	2022	2023
Index, fixed reference year	100	104	106.08
Index, previous year = 100	—	104	102
Annual change in %	—	4	2

```
year,cpi
2021,100.00
2022,104.00
```

File for the next example: income years 2021 and 2022. The base year is also 2021.

Fixed index: include the income years and the price base year.

Annual formats: include every intervening annual change. The app chains them.

An illustration in 2021 prices

Survey years 2022 and 2023. Incomes refer to the year before. Price base year 2021, with CPI 2021 = 100.

Survey / income year	CPI	Nominal mean	Mean in 2021 prices
2022 / 2021	100.00	10,400	10,400.00
2023 / 2022	104.00	10,920	10,500.00

Real welfare = current-price welfare × CPI(base) / CPI(income year)

Nominal mean rises 5.00%. Real mean rises 0.96%.
The app converts household observations before fitting the models.

Step 4: models on the chosen price basis

Price adjustment comes first. Log modelling is a separate decision.

Log + bias correction	Select log. Models use log of the arithmetic Direct area mean, with sampling variance V/Direct². Estimates return to currency units.
Bias correction	UFH: bc_sm uses emdi's Slud–Maiti correction. MFH: Conditional lognormal uses exp(eta + v/2). none omits correction. The target remains arithmetic mean welfare.
Other model choices	No transformation is an alternative. Review diagnostics and covariates. Use MFH2 by default and check PSU consistency and the covariance settings.

Zero and negative individual welfare stay in Direct. Log models require positive, finite area means.

Mean-welfare checks and outputs

Before running, check the Prices row in the Step 6 summary.

Output	Meaning / unit
Levels and RMSE	Currency in the selected constant prices
Changes and significance	Percentage change of the final mean estimates
Change-and-RMSE box plots	ln(m2/m1), without × 100: 0.00957 ≈ 0.96% change. RMSE uses the same log scale.
Excel columns ending in _eur	Changes and interval bounds: currency units. MSE: squared currency units. P-values: no units. The _eur suffix applies to any currency.

Zero variance or correlation at its limit: MFH change significance is provisional. Intervals may be too narrow.
Compare Direct and admissible models. More draws alone do not resolve this limitation.
UFH change intervals assume independent years. Review overlapping samples.

PART 04 · OPTIONAL

AI Assistant — Claude & ChatGPT

Acknowledge the external transfer first; use only an approved provider or gateway, then enter a key.



What the AI Assistant does

Plain-language interpretation in both HTML and Word reports, in your language.

Interpret diagnostics

Summarises convergence, normality and precision.

Review residual patterns

Reads Q–Q plots and descriptive normality checks.

Write report text

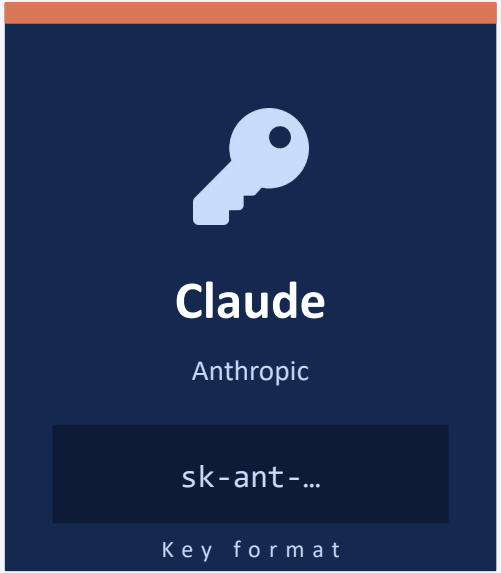
Labelled interpretation blocks in the HTML and Word reports.

Privacy Only aggregate tables are sent: no microdata and no domain names or IDs. Each AI block is marked as not yet reviewed; check it before sharing.

Get a Claude API key

Claude Console (Anthropic) — keys start with sk-ant-

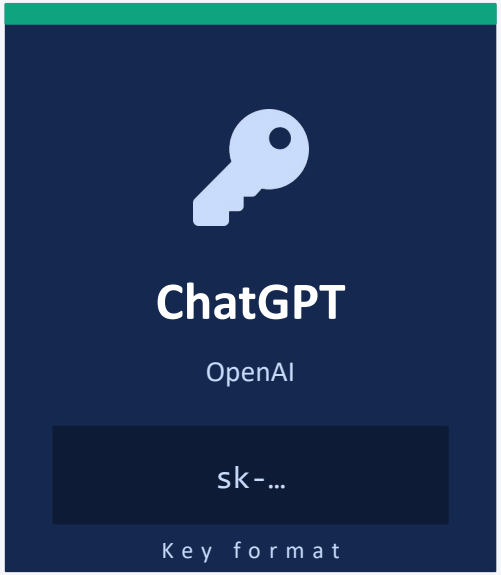
1. Open platform.claude.com — sign up or log in.
2. Settings → API keys — or go to platform.claude.com/settings/keys directly.
3. Create a new key — give it a name and copy it.
4. Add credit in Billing — pay-as-you-go; a Claude chat plan does not include API credit.
5. Tick both consent boxes in Step 5, then paste the key.



Get a ChatGPT (OpenAI) API key

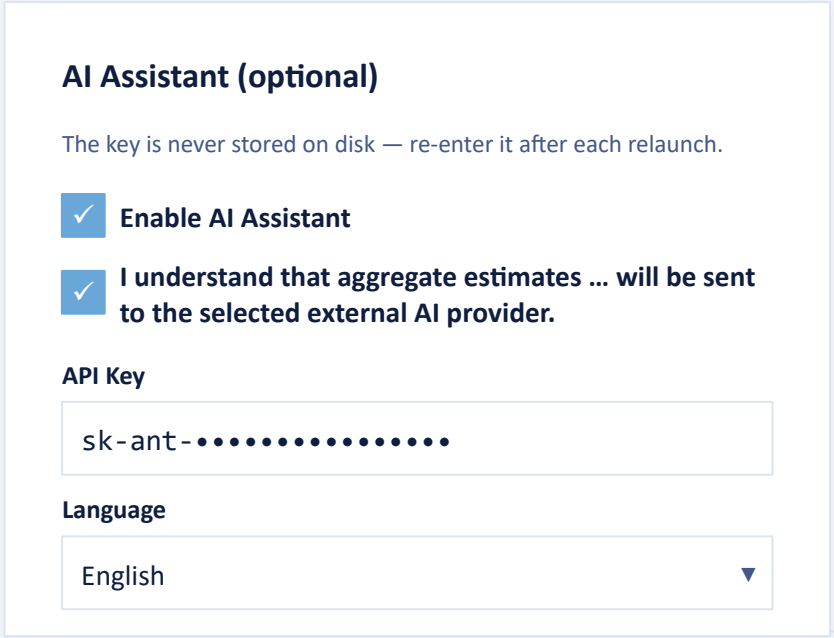
OpenAI's developer platform — keys start with sk-

1. Open platform.openai.com — sign up or log in.
2. Dashboard → API Keys — or go to platform.openai.com/api-keys directly.
3. Create a new key — give it a name and copy it.
4. Add credit in Billing — pay-as-you-go; a ChatGPT plan does not include API credit.
5. Tick both consent boxes in Step 5, then paste the key.



Enable the AI Assistant in the app

Two consent boxes, one key, one language — then you're ready.



1 **Enable AI and acknowledge the external transfer**

Both controls are required before key entry.

2 **Enter a Claude or OpenAI key**

Detected from the prefix (sk-ant-… or sk-…). Gateways: set SAE_*_BASE_URL before launch.

3 **Pick your language**

25 options. Every AI response comes back in that language.

25 languages supported

Every AI response is returned in the language you pick.

25

languages

24 EU official · + Arabic

Bulgarian	Croatian	Czech	Danish	Dutch
English	Estonian	Finnish	French	German
Greek	Hungarian	Irish	Italian	Latvian
Lithuanian	Maltese	Polish	Portuguese	Romanian
Slovak	Slovenian	Spanish	Swedish	Arabic

Troubleshooting

Quick fixes for the most common stumbling blocks.

Package errors on launch

Re-run the launcher you used. It reinstalls missing packages automatically.

Data file not selected or not found

Use Browse for each file in Step 1. Accepted formats are listed in Part 1.

Report or Word file missing

Check the run log, Pandoc, xml2 and zip; fix and regenerate. Word failure leaves HTML intact.

AI key or timeout errors

Paste only the key (no quotes or spaces); check billing. Timeouts: try again; see the run log.

Run Analysis won’t start

Click Check Data Readiness again after any change, and fix every ERROR line.

Geometry readiness error

IDs must match the survey and the mapped domain field. Shapefiles: a .zip with .shp, .shx, .dbf.

PRE-RELEASE FOR REVIEW

Review before circulation

Rerun selected country setups

```
EU_SAE_520_w5n.zip
```

Check fit and uncertainty

```
Review bootstrap diagnostics and sensitivity
```

Coverage is not certified

```
Boundary fits: change significance is provisional
```

Questions? Nobuo Yoshida · github.com/Noboyoshida227/eu-sae-application/issues